Principal Component Analysis (PCA) in 3D Shape Analysis

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Overview

- Introduction to Matrix Data Structure
- What is the PCA?
- How PCA Works
- What are Mesh and 3D Point Clouds Data?
- PCA for Feature Reduction of 3D Point Clouds
- Introduction to Course Work (3D Point Cloud Lumbar Vertebral Classification)
    - Goal of the Course Work
    - Dataset Details
    - Developing Pipeline
    - Your Report Summary

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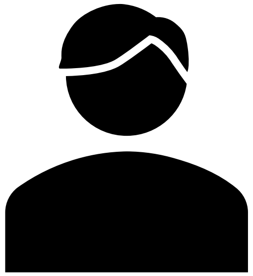
Matrix Data Structure

- What is a feature?
  - It is a measurable property of the data that we're trying to analyse.
  - For example, a person may be described by several features like age, gender, weight, height, and etc
  - The features are usually represented column-wise for each data



- Age: 26
- Gender: Male
- Height = 178 cm
- Weight = 86 Kg

$$\boldsymbol{x} = [\mathbf{26}, \mathbf{1}, \mathbf{178}, \mathbf{86}]$$

Matrix Data Structure

- We can gather all data with their features into a single matrix ($\boldsymbol{X}$)
  - $\boldsymbol{X} \in \mathbb{R}^{N \times M}$
  - N: Number of data
  - M: Number of features
- We may also have label or target matrix ($\boldsymbol{Y}$)
  - $\boldsymbol{Y} \in \mathbb{R}^{N}$

$\boldsymbol{X}$:

Age	Height	Weight	BMI	Smoking
68	189.5	64.6	18	0
58	168.5	89.6	31.6	0
44	160.9	86.7	33.5	1
72	161.8	85.8	32.8	0
37	165.2	64.2	23.5	0
50	164.3	74.3	27.5	1
68	149.1	62.2	28	0
48	182.6	66.3	19.9	0

$\boldsymbol{Y}$:

CVD Flag
1
0
1
1
1
0
1
1

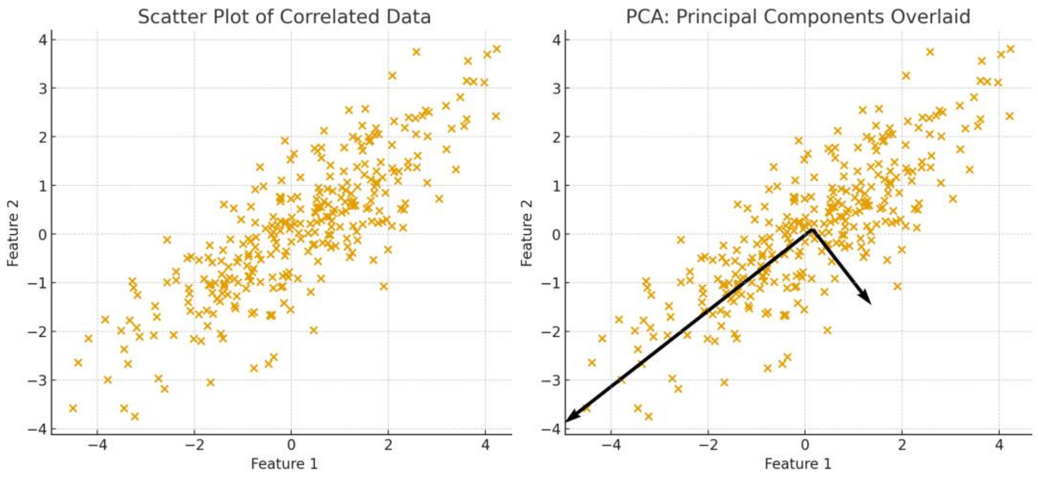
An example of feature matrix and label matrix representation. It is an artificial dataset for the people who have cardiovascular diseases or not.

What is PCA?

- In most data science scenarios, we have a large numbers of features.
  - It may lead our models to be overfitted, difficult visualisation and other issues.
  - Consequently, we need some techniques to reduce the number of features while retaining most of the desired information and get rid the irrelevant ones.
- **PCA is a one of these methods aiming to simplify a dataset by reducing the number features while preserving as much information as possible.**
  - It transforms the features of a dataset into a new feature set that are **orthogonal** and **uncorrelated**

How PCA works?

- PCA works by transforming the data into a new coordinate system that captures the most variance in the dataset.
- The new coordinate system =principal components = loading vectors
- By projecting the dataset on to these components, we can have new uncorrelated features for our dataset.



(a) (b)

Fig (a): The scatter points show the original 2D correlated data. Fig (b): The scatter plot of the data with the principal components that are shown by arrows.

How PCA works?

- Suppose we have $\boldsymbol{X} \in \mathbb{R}^{N \times M}$ as our feature matrix data, where N is the number of data and M is the number of features.
- We aim to have $\mathrm{R} \leq \min(\mathrm{N}, \mathrm{M})$ new uncorrelated and orthogonal features for our dataset using PCA
    - PCA Algorithm:
        1. Compute the mean of the feature matrix: μ
        2. Subtract the feature matrix from the mean vector: $\tilde{\mathbf{X}}$
        3. Compute the covariance matrix: $\mathbf{C}$
        4. Apply singular value decomposition (SVD) on the covariance matrix
            - $\mathbf{U} \in \mathbb{R}^{N \times \mathrm{R}}$ where $\mathbf{U}^{\boldsymbol{T}}\mathbf{U} = \mathbf{I}$
            - $\boldsymbol{S} \in \mathbb{R}^{\mathrm{R} \times \mathrm{R}}$ is the diagonal matrix of eigen values
            - $\mathbf{V}^{\boldsymbol{T}} \in \mathbb{R}^{\mathrm{R} \times \mathrm{M}}$ contains the principal components where $\mathbf{V}^{\boldsymbol{T}}\mathbf{V} = \mathbf{I}$
        5. Projecting the data to the principal components: $\mathbf{Z}$
- $\mathbf{Z}$ is the transformed dataset that has uncorrelated and orthogonal features

$$\mu = \frac{1}{N}\sum_{i=1}^{N} \mathbf{X}(i,:)$$

$$\tilde{\mathbf{X}} = \mathbf{X} - \mu$$

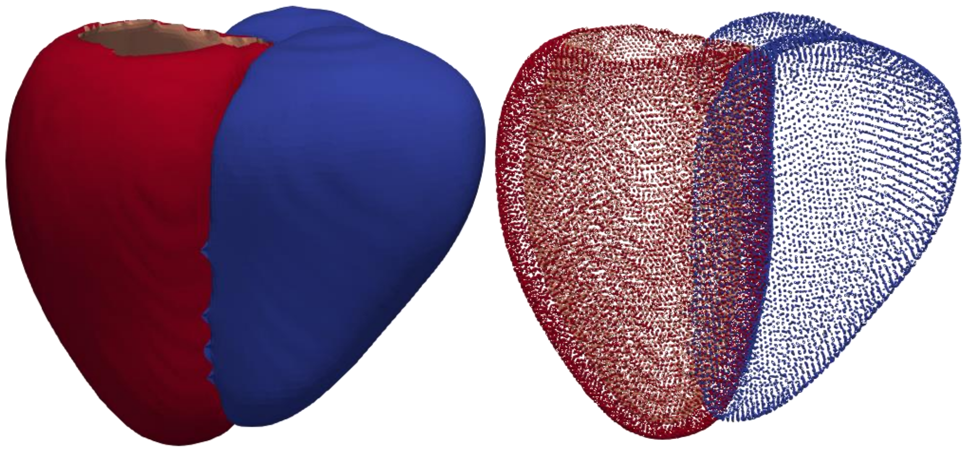
$$\mathbf{C} = \frac{1}{N-1}\tilde{\mathbf{X}}^T\tilde{\mathbf{X}}$$

$$\mathbf{USV}^T = \mathrm{SVD}(\mathbf{C})$$

$$\mathbf{Z} = \tilde{\mathbf{X}}\mathbf{V} \quad \text{where} \quad \mathbf{Z} \in \mathbb{R}^{N \times R}$$

What are 3D mesh and point cloud data?

- A 3D point cloud is a kind of vision data that contains a set of points with three coordinate values (x, y, z) representing the locations of points in 3D space.
- A 3D mesh is a collection of interconnected points that form a continuous surface
- The points of a mesh (a point cloud) can be represented by $\boldsymbol{X} \in \mathbb{R}^{N\times 3}$ where N is the number of points.
- Most of these data have a high number of features (3N) which negatively affects in data science applications.



(a) (b)

Fig (a): A sample of 3D heart mesh. Fig (b): A sample of 3D point cloud.

PCA for 3D point clouds

- Suppose we have L number of shapes $\boldsymbol{X_l} \in \mathbb{R}^{N\times3}$ $l = 1,2,\dots,\mathrm{L}$ and we gather all the flattened points array of the shapes into $\mathbf{H} \in \mathbb{R}^{\mathrm{L}\times3\mathrm{N}}$.
- If we apply PCA transformation on $\mathbf{H}$, what do the principal components resemble?
  - Regarding the descending order of the eigenvalues of principal components, they resemble the strongest pattern of variation across all point clouds.
  - For example:
    - The first component = the strongest pattern of variation across all point clouds
    - The second component = the second strongest, orthogonal to first one
- Using these components, we can project the point cloud dataset into a lower and more meaningful feature representation.

PCA for feature reduction of 3D point cloud data

- Given L number of shapes $\boldsymbol{X_l} \in \mathbb{R}^{N\times 3}$ $l = 1,2, \dots, \mathrm{L}$, we aim to use PCA to reduce the number of features of our point cloud data.
- Main algorithm:
  1. Flatten all the shapes and gather them into $\mathbf{H} \in \mathbb{R}^{\mathrm{L}\times 3\mathrm{N}}$.
  2. Apply PCA to $\mathbf{H}$ with the desired number of components ($\mathrm{R} \leq \min(\mathrm{L}, 3\mathrm{N})$).
  3. Project a shape ($\mathbf{Z} \in \mathbb{R}^{N\times 3}$) to the principal components $\mathbf{V} \in \mathbb{R}^{3N\times \mathrm{R}}$ using the following steps:
     1. Flattening $\mathbf{Z}$: $\boldsymbol{Z_f} \in \mathbb{R}^{3N}$
     2. Projecting $\boldsymbol{Z_f}$ to $\mathbf{V}$ by $\mathbf{Z}_\mathrm{p} = \boldsymbol{Z_f}\mathbf{V}$ where $\mathbf{Z}_\mathrm{p} \in \mathbb{R}^{\mathrm{R}}$

Shape Generation using PCA

- We also can generate some shapes using the mean vector of $\mathbf{H}, \widetilde{\mathbf{H}}$, and other obtained principal components using the following formulation:

$$\widetilde{\mathbf{H}} = \mathbf{1}/\boldsymbol{L} \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{L}} \mathbf{H}(\mathrm{i}, :)$$

$$\mathbf{Z}_{\boldsymbol{new}} = \widetilde{\mathbf{H}} + \alpha \mathbf{V}_{\boldsymbol{i}} \qquad -\mathbf{3}\sqrt{\boldsymbol{\lambda}_{\boldsymbol{i}}} < \alpha < \mathbf{3}\sqrt{\boldsymbol{\lambda}_{\boldsymbol{i}}}$$

$\mathbf{V}_{\boldsymbol{i}}$ is the *i*th principal component and $\boldsymbol{\lambda}_{\boldsymbol{i}}$ is its corresponding eigenvector

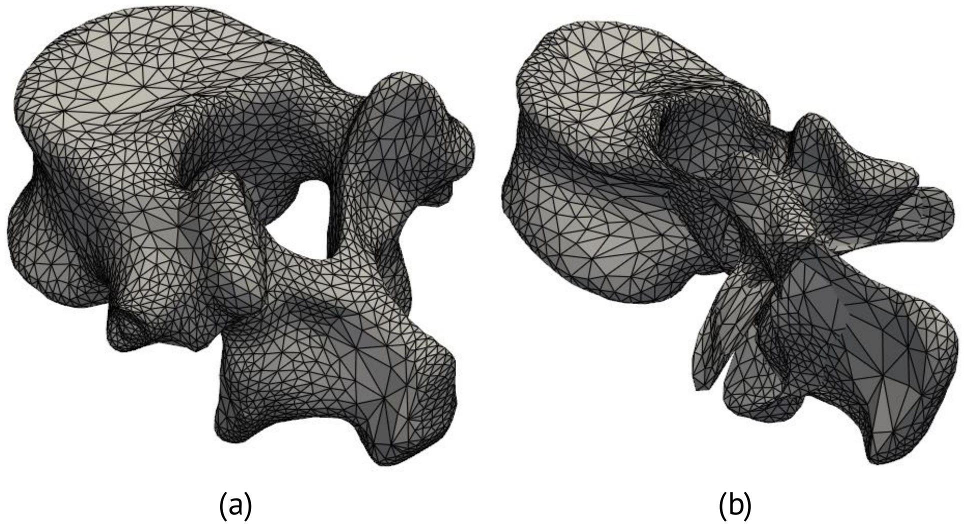
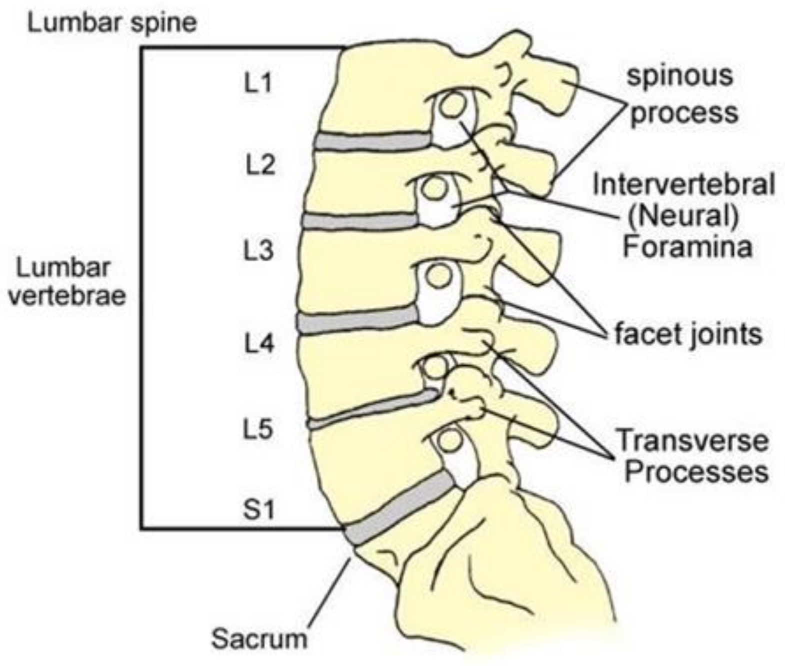


Fig (a): A sample of generated shapes using the first principal component. Fig (b): A sample of generated shapes using the second principal component.

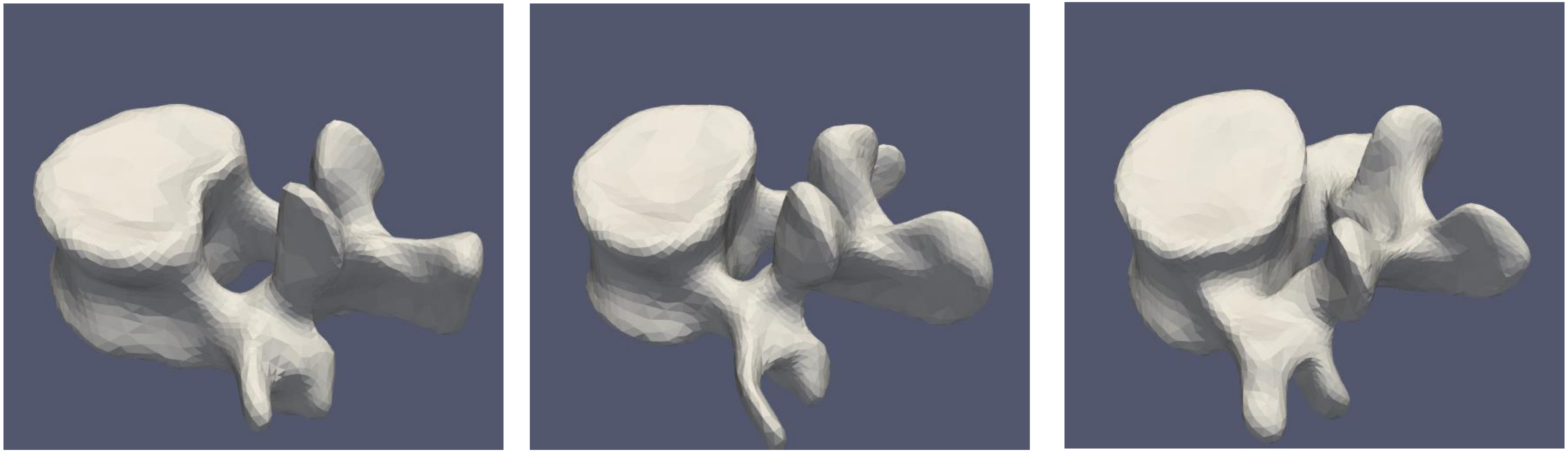
Introduction to task (3D Point Cloud Lumbar Parts Classification)

Goal of the Course Work and Dataset Details

- The lumbar region refers to the lower part of the spine and the lower back area of the body.
- It comprises five regions named L1, L2, L3, L4, and L5.
- The course of work
    - Developing a classification framework that classifies the 3D shapes of Lumbar regions to L1, L3, and L5 classes.
    - Dataset Details:
        - 90 Number of meshes (30 cases for each class)
        - Each mesh has 4000 points
        - All meshes have the corresponding points



Introduction to task (3D Point Cloud Lumbar Parts Classification)

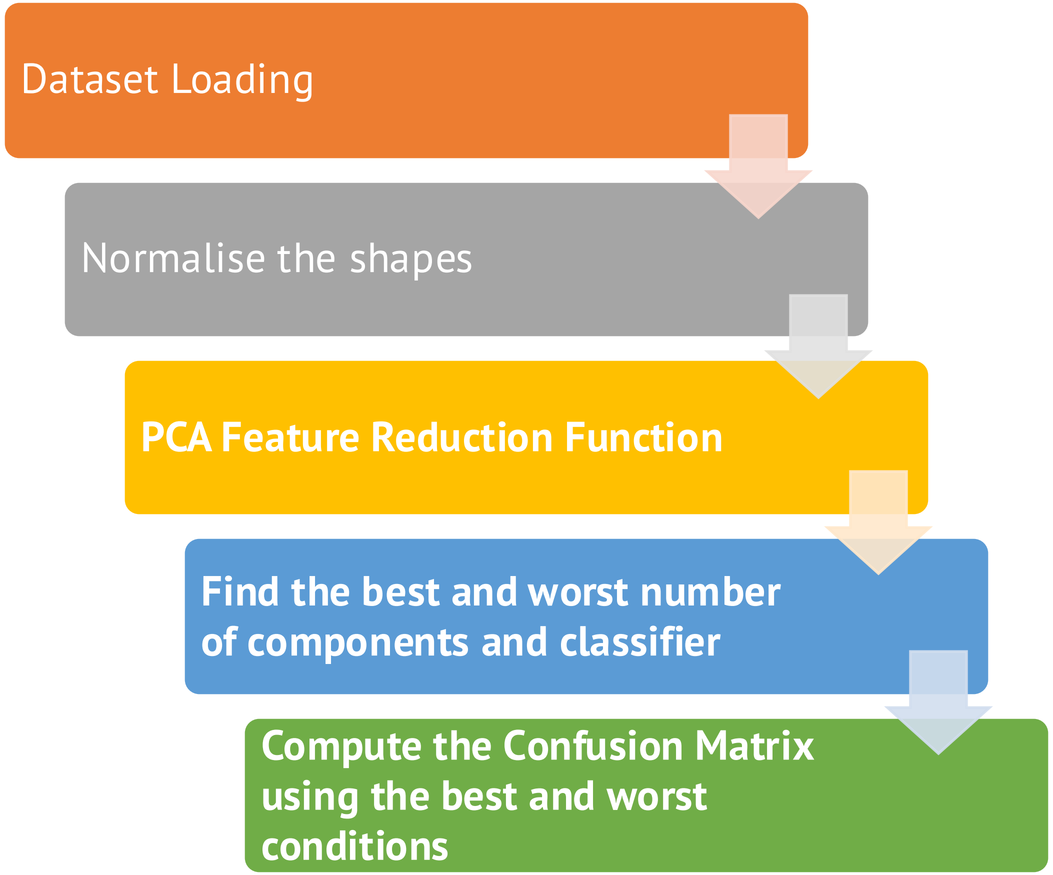


(a) (b) (c)

A sample of our dataset give one subject. Figs (a), (b), and (c) shows the L1, L3, and L5 shapes for the same subject, respectively.

Introduction to task (3D Point Cloud Lumbar Parts Classification)

Developing Pipeline



The flowchart of our classification framework

<u>Your Task:</u>

<u>6 members per group:</u>

- Write the PCA Feature reduction function
- Find the best and worst number of components and classifier
- Compute the Confusion Matrix using the best and worst conditions

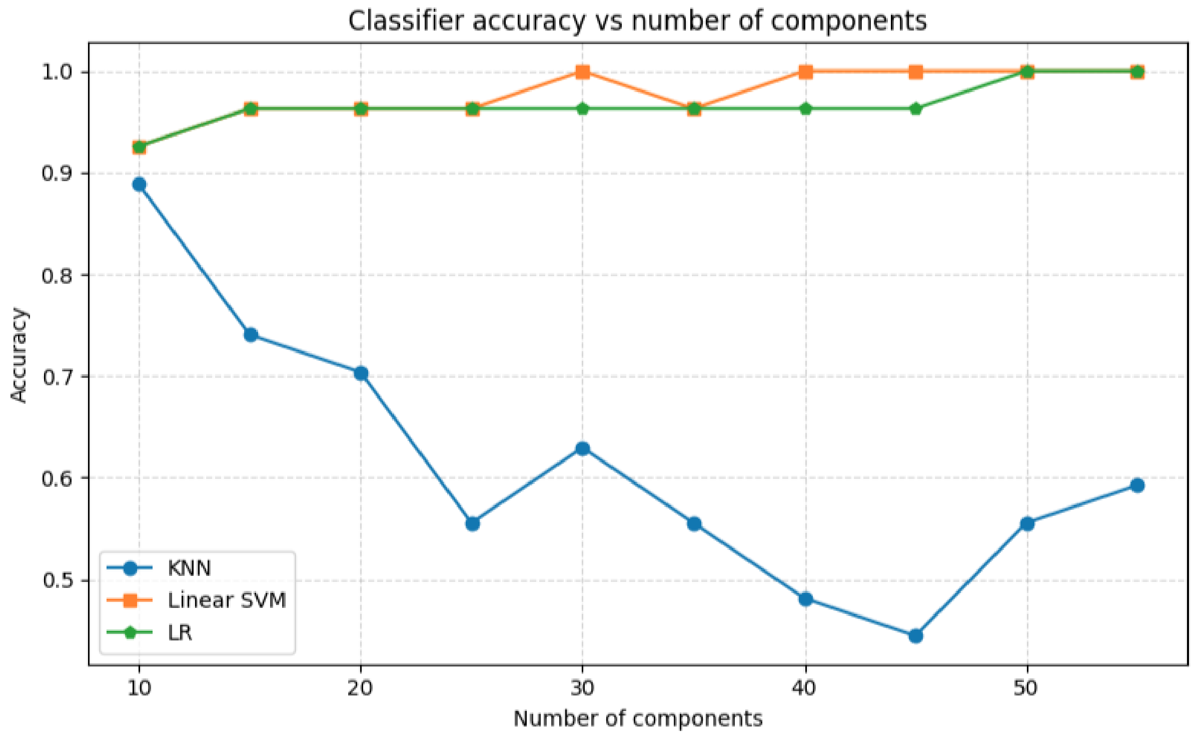
<u>4-6 members per group:</u>

- Write the PCA Feature reduction function
- Find the best and worst number of components and classifier

Introduction to task (3D Point Cloud Lumbar Parts Classification)

Report Summary

	Number of components			
Classifier	5	10	….	60
KNN	?	?	….	?
SVM	?	?	….	?
Logistic Regression	?	?	….	?



Example of the results report for finding the best and worst classifier and number of components

Introduction to task (3D Point Cloud Lumbar Parts Classification)

Report Summary

The submitted report should contain:

1. Introduction
    - Introduction to the case study
    - Clear statement of the purpose of the case study
2. Methods
    - Describe the pipeline
3. Results
    - Report the quantitative and qualitative results
4. Discussion
    - What conclusion can you draw from this case study

Thank you for your consideration